Computing the seasonal PM2.5 contrast

How does processing time change as the workload grows from 8 to 8,000 rows?

Method and workload

Eight observations from two sensors cover one January day and one June day. The heating-season mean exceeds the non-heating mean by 75.9 micrograms/m³. Repeating the rows gives four workloads with the same contrast. Each implementation ran five warm-up trials followed by 30 evaluation trials per size.

In-process latency by implementation

Rows	Row loop (ms)	Pandas (ms)	Speedup
8	0.587 ± 0.206	0.510 ± 0.252	1.15x
80	2.754 ± 0.186	0.265 ± 0.016	10.39x
800	26.862 ± 3.531	0.349 ± 0.164	77.05x
8,000	320.405 ± 90.569	0.555 ± 0.172	577.78x

Values are mean ± sample SD. Speedup is the ratio of unrounded means.

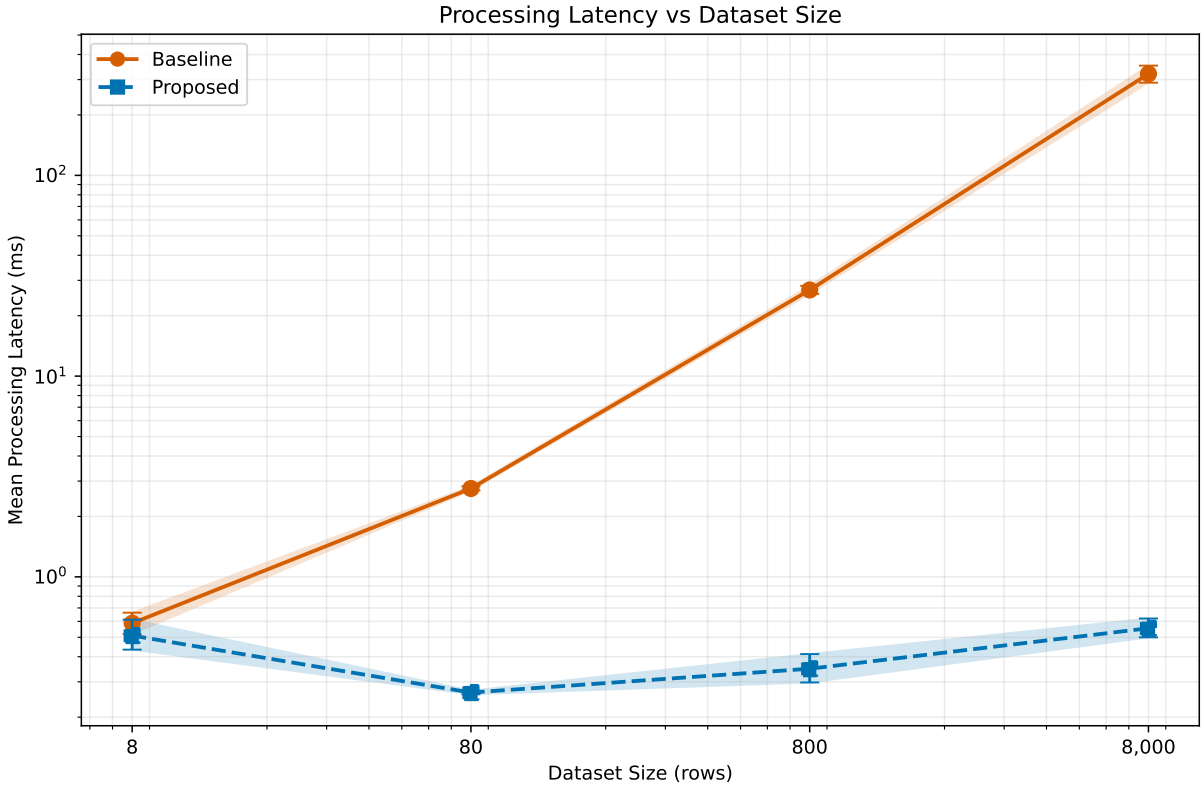

Figure 1. Logarithmic axes show the widening time gap. Bars and shaded bands show 95% bootstrap intervals for the mean. All 240 evaluation attempts succeeded.

Process time and interpretation

A separate experiment timed the complete Python program on the eight-row sample.

30 successful trials after five warm-up runs

A fresh process started for each trial. The timer includes imports, file reads and logging.

Mean 616.80 ms | SD 53.49 ms | Median 609.67 ms

P95 717.09 ms | P99 740.53 ms | Observed errors 0/30

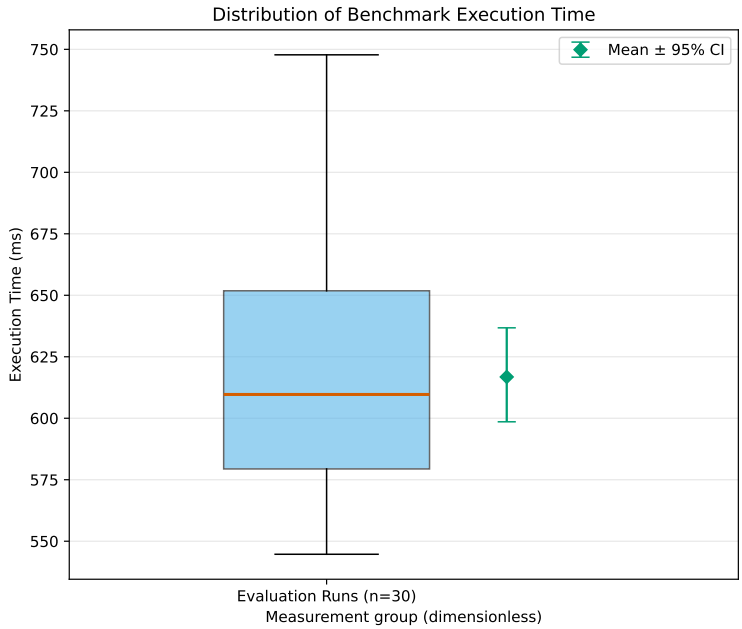

Figure 2. The box spans the middle half of individual runtimes. The diamond marks the mean and its 95% interval (598.56 to 636.77 ms, 10,000 resamples, seed 42).

What the measurements support

The Pandas implementation took less time on average at every size in this run. At eight rows, the mean intervals overlap. At 8,000 rows the gap is large, but both methods ran in fixed order on one machine. Order effects and the original hardware cannot be separated from the timings. P99 is especially sensitive to individual trials.

The original eight-row fixture has complete class separation at 35 micrograms/m³. Page 3 extends the seasonal comparison to observed city-days from 2021-2025. The domain model specifies the heating-by-calm interaction. Estimating it also requires wind measurements matched at the same time scale.

Open the tagged project, notebook and raw measurements on GitHub

Winters and summers in Almaty

Daily PM2.5 observations from the AirData.kz monitoring archive, 2021-2025.

Observed winter and summer means

Year	Days (W / S)	Winter mean	Summer mean	Difference
2021	90 / 92	84.45	11.65	72.81
2022	90 / 89	66.36	12.60	53.76
2023	90 / 92	57.65	14.39	43.26
2024	91 / 92	45.74	12.23	33.51
2025	90 / 92	48.68	9.84	38.84

Units: µg/m³. W = January, February and December of each year. S = June to August.

48.41 µg/m³ higher in winter

The pooled means were 60.54 across 451 winter days and 12.14 across 457 summer days. Each observed day has equal weight. All five yearly differences were positive. The snapshot covers 1,823 of 1,826 calendar days. June 9-11, 2022 are missing.

Benchmark on 908 distinct observed days

The loop and groupby returned the same contrast. Mean ± SD was 33.588 ± 6.145 ms for the loop and 1.010 ± 0.228 ms for groupby. Each method had five warm-ups and 30 evaluation trials, with alternating method order and zero observed errors.

Interpretation and data credit

Reported concentrations are city averages across geographic clusters. Coverage varied from 1 to 16 stations per day. Calendar season approximates heating exposure. The archive extract has no wind measurements. The difference cannot isolate effects of heating from weather or changes in the monitoring network.

Source: AirData.kz / Global Shapers Almaty Hub. Selected dates and columns. CC BY-NC 4.0.